LULC Entity Recognition and Relation Extraction Prompt

This appendix presents the complete prompt used for joint entity recognition and relation extraction in Land Use Land Cover (LULC) analysis. The prompt employs a Chain-of-Thought (CoT) + Few-Shot approach to guide the language model through systematic analysis of LULC-related text.
You are an expert in Land Use Land Cover (LULC) analysis. Perform joint entity recognition and relation extraction from the given sentence.
CHANGE: Verbs or noun phrases denoting measurable differences in land cover extent,quality, or characteristics between temporal observations. Captures both direction (increase,decrease, loss, gain) and nature (converted, degraded, improvement) of landscape dynamics.
• LOC: Proper nouns referring to geographically identifiable areas at any spatial scale (continents, countries, regions, cities, villages, named places).
• LULC: Nouns or noun phrases describing specific land surface types characterized by biophysical attributes or human utilization patterns (forests, woodlands, agricultural land,cropped fields).
• DATE: Numeric expressions or textual references specifying temporal points (1990, 2020,January 2000).
• PERCENT: Numerical expressions with percent symbols (%) or textual equivalents repre-senting proportional measurements.
• QUANTITY: Numerical values representing counts, ratios, indices, or non-percentage quantitative measures.
• COORDINATES: Numerical expressions representing geographic coordinates (40.7°N, 23.5°W).
• SURFACE_UNIT: Expressions combining numerical values with area measurement units
(km², hectares, acres).
• PROCESS: Nouns or verb phrases describing natural phenomena or anthropogenic activities
causing or resulting from LULC changes (droughts, desertification, cultivation, grazing).
• CARDINAL: Pure numerical values without specific units or context.
• TIME_PERIODS: Time spans with beginning and end points (1990-2000, between 1995 and 2005).
A.2.1 Entity Extraction Rules
Key extraction guidelines include:
1. Extract entities as concisely as possible (e.g., "loss" not "observed loss of woody vegetation")
2. List temporal references separately (e.g., "1970s" and "1980s" as distinct entities)
3. Classify "desertification" as PROCESS (the process of becoming desert), not LULC
4. Classify "woody vegetation" as LULC
A.3 Relation Type Definitions
The prompt restricts relation extraction to exactly 10 approved relation types:
CHANGE_TO, INCREASES_BY, DECREASES_BY, CAUSES, LOCATED_IN, OCCURS_DURING, MEASURES,
AFFECTS, FROM_TO, ENABLES
A.3.1 Key Relation Specifications

CHANGE_TO Indicates direct transformation between different LULC types only.
• Source: LULC, Target: LULC (different types)
• Correct: forest –CHANGE_TO– cropland
• Incorrect: forest –CHANGE_TO– forest
INCREASES_BY/DECREASES_BY Shows magnitude of change in land cover type.
• Source: LULC, Target: PERCENT or QUANTITY
• Example: built-up area –INCREASES_BY– 12.77%
CAUSES Establishes causal links between processes and changes.
LOCATED_IN Specifies spatial context for LULC, processes, or changes.
OCCURS_DURING Indicates temporal context for processes or changes.
MEASURES Quantifies entities through numerical values.
FROM_TO Represents value changes between quantities.

A.4 Chain-of-Thought Instructions

The prompt incorporates systematic reasoning steps:

1. Entity Identification: Systematically scan for all entity types
2. Relationship Analysis: Analyze "what happened to what, when, where, and why"
3. Validation: Verify each relation against approved relation types
4. Explicit Relations Only: Extract only explicitly stated relationships
5. CHANGE_TO Validation: Ensure connections only between different LULC types

A.4.1 Critical Thinking Rules

1. Verify actual transformations between different land types
2. Consider physical changes to land surface
3. Establish logical causality without meaningless loops
4. Recognize that percentages typically measure rather than cause changes
5. Apply temporal logic: changes occur during time periods
6. Apply spatial logic: entities are located in places

A.5 Few-Shot Examples

A.5.1 Example 1: Forest Conversion

Input: "Between 1995 and 2005, 30% of Brazil's forest area was converted to cropland due to agricultural expansion."
Chain-of-Thought Analysis:
THINKING:
Entity Identification:
- "Between 1995 and 2005" = TIME_PERIODS
- "30%" = PERCENT
- "Brazil" = LOC
- "forest area" = LULC
- "cropland" = LULC
- "converted" = CHANGE
- "agricultural expansion" = PROCESS

Relationship Analysis:
- Direct transformation between different LULC types → CHANGE_TO
- Percentage measures extent of change → MEASURES
- Process caused the conversion → CAUSES
- Spatial relationships → LOCATED_IN
- Temporal context → OCCURS_DURING

Output:
ENTITIES:

- Between 1995 and 2005 | TIME_PERIODS
- 30% | PERCENT
- forest area | LULC
- Brazil | LOC
- cropland | LULC
- converted | CHANGE
- agricultural expansion | PROCESS

RELATIONS:

- converted:CHANGE --OCCURS_DURING-- Between 1995 and 2005:TIME_PERIODS
- forest area:LULC --LOCATED_IN-- Brazil:LOC
- forest area:LULC --CHANGE_TO-- cropland:LULC
- 30%:PERCENT --MEASURES-- converted:CHANGE
- agricultural expansion:PROCESS --CAUSES-- converted:CHANGE

A.5.2 Example 2: Desertification Impact

Input: "Between 2000 and 2015, desertification in the Sahel region caused a significant loss of grassland."

Chain-of-Thought Analysis:

THINKING:

Entity Identification:

- "2000 and 2015" = TIME_PERIODS
- "desertification" = PROCESS
- "Sahel region" = LOC
- "loss" = CHANGE
- "grassland" = LULC

Relationship Analysis:

- Process directly caused change → CAUSES
- Change affects grassland → AFFECTS
- Process located in region → LOCATED_IN
- Change occurred during time period → OCCURS_DURING

Output:

ENTITIES:

- 2000 and 2015 | TIME_PERIODS
- desertification | PROCESS
- Sahel region | LOC
- loss | CHANGE
- grassland | LULC

RELATIONS:

- desertification:PROCESS --CAUSES-- loss:CHANGE
- loss:CHANGE --AFFECTS-- grassland:LULC
- desertification:PROCESS --LOCATED_IN-- Sahel region:LOC
- loss:CHANGE --OCCURS_DURING-- 2000 and 2015:TIME_PERIODS

Output:

ENTITIES:

- Urban area | LULC
- expanded | CHANGE
- 30 km^2 | SURFACE_UNIT
- 1990 and 2020 | TIME_PERIODS
- 45 km^2 | SURFACE_UNIT

- Shanghai | LOC
RELATIONS:
- expanded:CHANGE --LOCATED_IN-- Shanghai:LOC
- 30 km²:SURFACE_UNIT --FROM_TO-- 45 km²:SURFACE_UNIT
- 30 km²:SURFACE_UNIT --MEASURES-- Urban area:LULC
- 45 km²:SURFACE_UNIT --MEASURES-- expanded:CHANGE

A.6 Output Format Specification

The prompt requires strict adherence to the following output format:
ENTITIES:
- entity_text | ENTITY_TYPE
RELATIONS:
- entity:ENTITY_TYPE --RELATIONSHIP-- entity:ENTITY_TYPE

Output Requirements
1. Extract only explicitly stated or directly implied relations
2. Include entity labels in the specified format
3. Provide confidence levels (HIGH/MEDIUM/LOW) for each relation
4. Avoid creating relations between same-type entities using CHANGE_TO
5. Maintain conciseness and accuracy

A.7 Prompt Design Rationale

This prompt design combines Chain-of-Thought reasoning with few-shot examples to ensure:
- Systematic Analysis: Step-by-step entity identification and relationship analysis
- Consistency: Strict entity and relation type constraints prevent inconsistent outputs
- Transparency: Explicit reasoning traces enable validation and debugging
- Domain Expertise: LULC-specific rules and examples guide appropriate extractions
- Robustness: Multiple examples demonstrate handling of diverse sentence structures